

INHA UNIVERSITY TASHKENT
DEPARTMENT OF CSE & ICE
SPRING SEMESTER 2024
SOC 2040 - SYSTEMS PROGRAMMING
LAB ASSIGNMENT 2

INSTRUCTIONS :

- ALL LAB ASSIGNMENTS ARE TO BE COMPLETED IN GROUPS OF 4 TO 6 STUDENTS.
- LAB ASSIGNMENT REPORT SHOULD BE PREPARED USING THE LAB ASSIGNMENT2 REPORT TEMPLATE PROVIDED AT THE ECLASS PORTAL.
- **ONE HARD COPY OF THE LAB ASSIGNMENT2 REPORT OF EACH GROUP (TEAM LEADER'S REPORT) IN SPIRAL BOUND FORM MUST BE HANDED IN BY THE TEAM LEADER BY VISITING MY OFFICE ON TUESDAY, 26TH MARCH 2024 ANYTIME BETWEEN 2PM AND 5PM.**
- EVERY MEMBER OF THE TEAM MUST UPLOAD THE SOFTCOPY OF THE REPORT AT THE E- CLASS PORTAL.
- **LAST DATE FOR SUBMISSION OF THE LAB ASSIGNMENT2 REPORT AT THE ECLASS PORTAL IS MONDAY 25TH MARCH 2024.**
- LATE SUBMISSIONS ARE NOT ENTERTAINED, ADHERE TO THE DEADLINE STRICTLY
- READ THE QUESTIONS CORRECTLY & CAREFULLY

FOR ALL PART-A : PRACTICE QUESTIONS, ALL THE STEPS INVOLVED IN ARRIVING AT THE SOLUTION SHOULD BE HANDWRITTEN IN THE REPORT. OTHERWISE A GRADE 0 WILL BE GIVEN.

PLEASE NOTE THAT FOR ALL THE QUESTIONS IN PART-A, ANSWERS MUST BE HANDWRITTEN BY EACH MEMBER ON A4 SIZE WHITE PAPER (USED IN THE REPORT).

- EVERY MEMBER OF THE TEAM MUST SOLVE ALL THE QUESTIONS OF PART-A Q1) TO Q13). IT MUST BE HANDWRITTEN USING THE LAB ASSIGNMENT 2 REPORT TEMPLATE.**
- ANSWERS CAN BE WRITTEN ON BOTH THE SIDES OF THE PAPER AND PRINTOUTS CAN BE TAKEN ON BOTH SIDES OF THE PAPER.**
- ANSWERS HANDWRITTEN MUST BE LEGIBLE AND PERFECTLY READABLE. SIMILARLY PRINTOUTS TAKEN MUST BE LEGIBLE AND PERFECTLY READABLE – OTHERWISE IT WILL NOT BE GRADED AND GIVEN A GRADE 0 OUTRIGHTLY.**
- AS A TEAM WORK, PARTS IN EACH QUESTION IN PART-A SHOULD BE SHARED AMONG THE TEAM MEMBERS AS INDICATED IN EVERY QUESTION.**
- IN EACH QUESTION, IT IS CLEARLY INDICATED WHICH PARTS OF THE QUESTION EVERY MEMBER HAS TO SOLVE.**
- LATER ALL MEMBERS SHOULD SHARE THEIR SOLUTIONS IN ORDER TO PUT THE SOLUTIONS IN THEIR RESPECTIVE REPORTS. THE SCAN OF ALL THE ANSWERS OF PART-A CAN BE INCLUDED IN**

THE LAB ASSIGNMENT 2 REPORT SOFTCOPY (TEMPLATE PROVIDED).

- **TEAM LEADER MUST SUBMIT THE HARDCOPY OF HIS/HER REPORT ALONG WITH THE ORIGINAL HANDWRITTEN SHEETS OF SOLVED QUESTIONS RECEIVED FROM ALL THE MEMBERS FOR PART A - 13 QUESTIONS. WITHOUT THESE GRADE 0 WILL BE GIVEN TO ALL MEMBERS.**
- **IN THE LAB REPORT HARD COPY AND SOFT COPY, IT MUST BE CLEARLY WRITTEN WHO HAS SOLVED WHICH PARTS OF EACH QUESTION BY WRITING YOUR NAME, STUDENT ID & SECTION NO.**
- **FOR PART-B : C PROGRAMMING QUESTIONS Q14) TO Q20) - PROGRAM DESIGN AND PROGRAM WRITING SHOULD BE SHARED AMONG THE MEMBERS BY THE TEAM LEADER.**
- **ALL PROGRAMS WRITTEN MUST BE EXECUTED BY ALL TEAM MEMBERS INCLUDING TEAM LEADER ON THEIR RESPECTIVE LAPTOPS ON LINUX SYSTEM AND SCREENSHOTS OF THE PROGRAM ENTERED, INPUTS GIVEN AND RESULTS OBTAINED ON THEIR LAPTOP MUST BE TAKEN.**
- **EACH TEAM MEMBER SHOULD TAKE 6 OR MORE DIFFERENT INPUTS AND EXECUTE EVERY PROGRAM ON THEIR LAPTOP.**

SCREENSHOTS OF THESE INCLUDING THE RESULTS CAN BE SHARED WITH OTHER

MEMBERS OF THE TEAM SO THAT LAB REPORT SUBMITTED BY EACH MEMBER WILL HAVE ANSWERS FOR ALL THE QUESTIONS IN THE LAB ASSIGNMENT2 – THAT IS, 36 OUTPUTS FOR 36 DIFFERENT INPUTS FOR EACH PROGRAMMING QUESTION.

- IN THE TEAM LEADER'S HARD COPY AND ALL MEMBERS INCLUDING TEAM MEMBER'S SOFT COPY OF THE LAB REPORT , FOR EACH C PROGRAMMING QUESTION IN PART-B, YOU NEED TO PROVIDE ALGORITHM, PROGRAM AND SCREENSHOTS OF PROGRAM ENTERED, INPUTS GIVEN & RESULTS OBTAINED.

- ALL FILES CONTAINING C PROGRAMS WRITTEN AND EXECUTED FOR ALL THE PART-B QUESTIONS SHOULD BE UPLOADED AT THE E-CLASS PORTAL ALONG WITH THE LAB ASSIGNMENT 2 REPORT.

- YOU ARE REQUIRED TO PUT ALL THE FILES IN A FOLDER AND GIVE THE FOLDER NAME - YOUR ID FOLLOWED BY SPLABASSIGNMENT2 (EXAMPLE: U2210101_SPLABASSIGNMENT2) AND UPLOAD AT THE ECLASS PORTAL.**

- LAST DATE FOR SUBMISSION OF THE LAB ASSIGNMENT2 REPORT AT THE ECLASS PORTAL IS MONDAY 25TH MARCH 2024.**
- LATE SUBMISSIONS ARE NOT ENTERTAINED, ADHERE TO THE DEADLINE STRICTLY**
- READ THE QUESTIONS CORRECTLY & CAREFULLY**

QUESTIONS :

PART – A : PRACTICE QUESTIONS

NO MARKS WILL BE GIVEN TO PART-A QUESTIONS IF ONLY ANSWER IS WRITTEN WITHOUT SHOWING COMPLETE WORKING STEPS TO ARRIVE AT THE SOLUTION.

IN ORDER TO GET FULL MARKS, ALL THE WORKING STEPS TO ARRIVE AT THE SOLUTION SHOULD BE SHOWN.

Q1) Convert each of the following binary numbers to

- (i) Unsigned form - B2U**
- (ii) Signed Magnitude form - B2S**
- (iii) One's Complement form - B2O**
- (iv) Two's Complement form - B2T**
- (v) Octal form - B2OCT**
- (vi) Hexadecimal form - B2HEX**

SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THESE QUESTIONS

- a) 11110001**
- b) 1110011011110110010**
- c) 01110110001110111000110000100111**

MEMBER2 HAS TO SOLVE THESE QUESTIONS

- d) 10100110
- e) 1110010111011011
- f) 110101111011101111101101

MEMBER3 HAS TO SOLVE THESE QUESTIONS

- g) 0100011011
- h) 10101010111010
- i) 10101111010101111111101110111110

MEMBER4 HAS TO SOLVE THESE QUESTIONS

- j) 11111111111
- k) 01110001011110110111
- l) 0111001100101011

MEMBER5 HAS TO SOLVE THESE QUESTIONS

- m) 011100110101
- n) 1100111010101010
- o) 10010101101001111101011011001011

MEMBER6 HAS TO SOLVE THESE QUESTIONS

- p) 101010010110
- q) 01110011001101
- r) 11111111111111111111111111111111

FOR A TEAM OF 5 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS p), q), r) ARE TO BE DONE BY MEMBER1- p), MEMBER 2- q), & MEMBER3- r)

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS m), n), o), p), q) r) ARE TO BE DONE BY MEMBER1- m), n), MEMBER2 -o), MEMBER3 – p), q), & MEMBER4 – r)

Q2) Convert the following Decimal numbers to

- (i) Binary form,**
- (ii) Octal form and**
- (iii) Hexadecimal form**

SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THESE QUESTIONS

- a) 78 b) 8476 c) 99999**

MEMBER2 HAS TO SOLVE THESE QUESTIONS

- d) 189 e) 5492 f) 66678**

MEMBER3 HAS TO SOLVE THESE QUESTIONS

- g) 234 h) 9428 i) 40899**

MEMBER4 HAS TO SOLVE THESE QUESTIONS

- j) 656 k) 3697 l) 27895**

MEMBER5 HAS TO SOLVE THESE QUESTIONS

- m) 763 n) 6724 o) 17835**

MEMBER6 HAS TO SOLVE THESE QUESTIONS

- p) 989 q) 2589 r) 32567**

FOR A TEAM OF 5 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS p), q) r) ARE TO BE DONE BY MEMBER2- p), MEMBER 3- q), MEMBER4- r)

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS m), n), o), p), q) r) ARE TO BE DONE BY

**MEMBER1- m), n), MEMBER2 -o), MEMBER3 – p), q),
MEMBER4 – r)**

Q3) Convert the following real numbers in decimal form to Binary form (provide atleast 16 bits after the binary point)

**SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE
WHITE PAPERS**

**FOR A TEAM OF 6 MEMBERS – MEMBER WISE
QUESTIONS ALLOTTMENT IS AS FOLLOWS:**

MEMBER1 HAS TO SOLVE THESE QUESTIONS

a) 78.895 b) 2589.22 c)99999.666663333

MEMBER2 HAS TO SOLVE THESE QUESTIONS

d)189.30 e) 8476.353 f) 27895.9841875

MEMBER3 HAS TO SOLVE THESE QUESTIONS

g)234.978 h) 6724.967 i) 66678.7778889

MEMBER4 HAS TO SOLVE THESE QUESTIONS

j) 656.656 k) 9428.444 l) 17835.7689

MEMBER5 HAS TO SOLVE THESE QUESTIONS

m) 763.3774 n) 5492.745 o)40899.123456

MEMBER6 HAS TO SOLVE THESE QUESTIONS

p)989.9999 q) 3697.45 r) 82567.78357

**FOR A TEAM OF 5 MEMBERS – THEY NEED TO
FOLLOW THE SAME EXCEPT THAT ADDITIONAL
QUESTIONS p), q) r) ARE TO BE DONE BY MEMBER3-
p), MEMBER 4- q), MEMBER1- r)**

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS m), n), o), p), q) r) ARE TO BE DONE BY MEMBER1- m), n), MEMBER2 -o), MEMBER3 – p), q), MEMBER4 – r)

Q4) Convert the following real numbers in binary form to Decimal form.

SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THESE QUESTIONS

- a) 00110011.10101010 b) 1110111010.10110101
c) 1010011110111110110110111111001.0011001101110111

MEMBER2 HAS TO SOLVE THESE QUESTIONS

- d) 10101110.01110111 e) 1101100101110111.000011111101
f) 01101111011101011010111010111101.0111110000011110

MEMBER3 HAS TO SOLVE THESE QUESTIONS

- g) 11110101.1100110011 h) 1101101111011011.111101111
i) 11111111111111111111111111111111.1011011011110110

MEMBER4 HAS TO SOLVE THESE QUESTIONS

- j) 01111111.01010101 k) 100010100110.00011101
l) 011110111101011011011101101100.11111011111

MEMBER5 HAS TO SOLVE THESE QUESTIONS

- m) 1001001011.0111001111 n) 0111110110101.10110110011101111
o) 101010100101011011.11011101

MEMBER6 HAS TO SOLVE THESE QUESTIONS

- p) 111111111.11111111 q) 010111101111.00111001111
r) 01010110101011110.110110110110

FOR A TEAM OF 5 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS p), q) r) ARE TO BE DONE BY MEMBER1- p), MEMBER 2- q), MEMBER4- r)

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS m), n), o), p), q) r) ARE TO BE DONE BY MEMBER1- m), n), MEMBER2 -o), MEMBER3 – p), q), MEMBER4 – r)

Q5) Convert the following Hexadecimal numbers to

- (i) Binary form,**
- (ii) Decimal form and**
- (iii) Octal form**

SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THESE QUESTIONS

a) 0xEF CB b) 0x1A2B3C4D5E c) 0x6ABCDEFFEDCBA6

MEMBER2 HAS TO SOLVE THESE QUESTIONS

d) 0x7BAD e) 0xBAD789BAC f) 0x987654ABCDEF3210

MEMBER3 HAS TO SOLVE THESE QUESTIONS

g) 0xABBA h) 0xFADEDAD i) 0xFEEDBAC3210CABADA

MEMBER4 HAS TO SOLVE THESE QUESTIONS

j) 0xbabadad k) 0xABCDEF9876 l) 0xBADDEED98765BEED

MEMBER5 HAS TO SOLVE THESE QUESTIONS

m) 0xD2E3F4 n) 0x9F8E7D6C5D o) 0x4BAD6CAB8DAD789B

MEMBER6 HAS TO SOLVE THESE QUESTIONS

p) 0xA9B8C7 q) 0xCADDEAD r) 0x123456789FEDCBAD

FOR A TEAM OF 5 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS p), q) r) ARE TO BE DONE BY MEMBER2- p), MEMBER 3- q), MEMBER4- r)

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS m), n), o), p), q) r) ARE TO BE DONE BY MEMBER1- m), n), MEMBER2 -o), MEMBER3 – p), q), MEMBER4 – r)

Q6) Convert the following Octal numbers to

- (i) Binary form,**
- (ii) Decimal form and**
- (iii) hexadecimal form**

SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THESE QUESTIONS

a) 17761 b) 77126644 c) 7766554433221100

MEMBER2 HAS TO SOLVE THESE QUESTIONS

d) 67453 e) 543212345 f) 1122334455667700

MEMBER3 HAS TO SOLVE THESE QUESTIONS

g) 23467 h) 56677374 i) 1234567654321010

MEMBER4 HAS TO SOLVE THESE QUESTIONS

j) 764523

k) 4433227766

l) 7654321012345670

MEMBER5 HAS TO SOLVE THESE QUESTIONS

m) 467426

n) 26745412

o) 753642532355452

MEMBER6 HAS TO SOLVE THESE QUESTIONS

p) 777666

q) 765432167

r) 456701235766777

FOR A TEAM OF 5 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS p), q) r) ARE TO BE DONE BY MEMBER1- p), MEMBER 3- q), MEMBER4- r)

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS m), n), o), p), q) r) ARE TO BE DONE BY MEMBER1- m), n), MEMBER2 -o), MEMBER3 – p), q), MEMBER4 – r)

Q7) Convert the following decimal numbers into the IEEE 754 format for single precision floating point numbers and express each in packed hex form :

SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THESE QUESTIONS

a) 78.895

b) -9428.444

c) 99999.666663333

MEMBER2 HAS TO SOLVE THESE QUESTIONS

d) 189.30 e) 8476.353 f) - 66678.7778889

MEMBER3 HAS TO SOLVE THESE QUESTIONS

g) -234.978 h) -6724.967 i) -40899.123456

MEMBER4 HAS TO SOLVE THESE QUESTIONS

j) 656.656 k) 5492.745 l) 27895.9841875

MEMBER5 HAS TO SOLVE THESE QUESTIONS

m) 763.3774 n) -3697.45 o) -17835.7689

MEMBER6 HAS TO SOLVE THESE QUESTIONS

p) -989.9999 q) 2589.22 r) 12567.78357

FOR A TEAM OF 5 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS p), q) r) ARE TO BE DONE BY MEMBER2- p), MEMBER 3- q), MEMBER4- r)

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS m), n), o), p), q) r) ARE TO BE DONE BY MEMBER1- m), n), MEMBER2 -o), MEMBER3 – p), q), MEMBER4 – r)

Q8) Convert the following binary numbers into the IEEE 754 format for single precision floating point numbers and express each in packed hex form :

SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THESE QUESTIONS

- a) 00110011.10101010
- b) 0111110110101.1011011001110111
- c) 01101111011101011010111010111101.0111110000011110

MEMBER2 HAS TO SOLVE THESE QUESTIONS

- d) 10101110.01110111
- e) 11111111111111111111111111111111.10110110110110
- f) 11110101.1100110011

MEMBER3 HAS TO SOLVE THESE QUESTIONS

- g) 01111111.01010101
- h) 1001001011.0111001111
- i) 01111011110101101100.111111011111

MEMBER4 HAS TO SOLVE THESE QUESTIONS

- j) 1111111111.11111111
- k) 010111101111.00111001111
- l) 01010110101011110.110110110110

MEMBER5 HAS TO SOLVE THESE QUESTIONS

- m) 1110111010.10110101
- n) 100010100110.00011101
- o) 101010100101011011.11011101

MEMBER6 HAS TO SOLVE THESE QUESTIONS

- p) 1101101111011011.111101111
- q) 1101100101110111.000011111101
- r) 10100111101111110110110111111001.0011001101110111

FOR A TEAM OF 5 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS p), q) r) ARE TO BE DONE BY MEMBER1- p), MEMBER 2- q), MEMBER3- r)

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS m), n), o), p), q) r) ARE TO BE DONE BY

**MEMBER1- m), n), MEMBER2 -o), MEMBER3 – p), q),
MEMBER4 – r)**

Q9) Convert the following binary numbers into the IEEE 754 format for DOUBL PRECISION floating point numbers and express each in packed hex form :

SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THESE QUESTIONS

- a) 1001101101111011.1010101010101010
- b) 1011111011111101101.1011011001001001101111
- c) 101101111011101011010111011110111101.0111110000011100010

MEMBER2 HAS TO SOLVE THESE QUESTIONS

- d) 10100011101110.01110110001111000111001110001
- e) 1111111111111111111111111111.10110110110110110110
- f) 11000111100101010111000101100110101.1100110011001100110011

MEMBER3 HAS TO SOLVE THESE QUESTIONS

- g) 0111110000001111.010101011100110011010101
- h) 10010010111100011110011011.01110011110100111011
- i) 01111011110111111101000000101101110.11111011111000111110111

MEMBER4 HAS TO SOLVE THESE QUESTIONS

- j) 1111111111111111.11111111111111111111
- k) 010111101111101111111011011.001110111110111001
- l) 010101101010111010101010101010111110.110110110110111010101

MEMBER5 HAS TO SOLVE THESE QUESTIONS

- m) 1110111001100110100110010.10110111111011101100101
- n) 10001101011110100110.0001111100110001110001111001
- o) 101010100101011011.11011101

MEMBER6 HAS TO SOLVE THESE QUESTIONS

p) 1101101111011011.111101111

q) 1101100101110111.00001111101

r) 10100111101111101101011011111001.0011001101110111

FOR A TEAM OF 5 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS p), q) r) ARE TO BE DONE BY MEMBER2- p), MEMBER 3- q), MEMBER4- r)

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS m), n), o), p), q) r) ARE TO BE DONE BY MEMBER1- m), n), MEMBER2 -o), MEMBER3 – p), q), MEMBER4 – r)

Q9) [A] Given the following real numbers, Express them in IEEE 754 single precision floating point forms

SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THESE QUESTIONS

a) N1= 2589.22

b) N2 = -3697.45

MEMBER2 HAS TO SOLVE THESE QUESTIONS

c) N3 = 5492.745

d) N4 = -6724.967

MEMBER3 HAS TO SOLVE THESE QUESTIONS

e) N5 = 8476.353

f) N6 = -9428.444

MEMBER4 HAS TO SOLVE THESE QUESTIONS

g) $N7 = 12567.78357$

h) $N8 = -17835.7689$

MEMBER5 HAS TO SOLVE THESE QUESTIONS

i) $N9 = 27895.9841875$

j) $N10 = -40899.123456$

MEMBER6 HAS TO SOLVE THESE QUESTIONS

k) $N11 = -66678.7778889$

l) $N12 = -989.9999$

FOR A TEAM OF 5 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS k), l) ARE TO BE DONE BY MEMBER1-k), MEMBER 2- l))

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS i), j), k), l), ARE TO BE DONE BY MEMBER1- i), MEMBER2 -j), MEMBER3 – k), MEMBER4 – l)

[B] Perform the following operations and express the results in packed hex form :
SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THESE QUESTIONS

(i) $N1 + N2$

(ii) $N1 + N12$

MEMBER2 HAS TO SOLVE THESE QUESTIONS

(iii) $N2 + N4$

(iv) $N3 + N11$

MEMBER3 HAS TO SOLVE THESE QUESTIONS

(v) $N5 + N6$

(vi) $N6 + N10$

MEMBER4 HAS TO SOLVE THESE QUESTIONS

(vii) $N7 + N8$

(viii) $N8 + N9$

MEMBER5 HAS TO SOLVE THESE QUESTIONS

(ix) $N9 + N11$

(x) $N8 + N4$

MEMBER6 HAS TO SOLVE THESE QUESTIONS

(xi) $N6 + N11$

(xii) $N7 + N12$

FOR A TEAM OF 5 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS xi), xii) ARE TO BE DONE BY MEMBER3- xi), MEMBER 4- xii))

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS ix), x), xi), xii), ARE TO BE DONE BY MEMBER1- ix), MEMBER2 - x), MEMBER3 – xi), MEMBER4 – xii)

Q10) [A] Given the following real numbers, Express them in IEEE 754 single precision floating point forms :

SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THESE QUESTIONS

a) N1= 2589.22

b) N2 = -3697.45

MEMBER2 HAS TO SOLVE THESE QUESTIONS

c) N3 = 5492.745

d) N4 = -6724.967

MEMBER3 HAS TO SOLVE THESE QUESTIONS

e) N5 = 8476.353

f) N6 = -9428.444

MEMBER4 HAS TO SOLVE THESE QUESTIONS

g) N7 = 12567.78357

h) N8 = -17835.7689

MEMBER5 HAS TO SOLVE THESE QUESTIONS

i) N9 = 27895.9841875

j) N10 = -40899.123456

MEMBER6 HAS TO SOLVE THESE QUESTIONS

k) N11 = - 66678.7778889

l) N12 = -989.9999

FOR A TEAM OF 5 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS k), l) ARE TO BE DONE BY MEMBER1-k), MEMBER 2- l))

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS i), j), k), l), ARE TO BE DONE BY MEMBER1- i), MEMBER2 -j), MEMBER3 – k), MEMBER4 – l)

[B] Perform the following operations and express the results in packed hex form :

SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THESE QUESTIONS

- (i) N1 - N2 (ii) N12 – N1

MEMBER2 HAS TO SOLVE THESE QUESTIONS

- (iii) N2 - N4 (iv) N3 - N11

MEMBER3 HAS TO SOLVE THESE QUESTIONS

- (iv) N6 - N5 (vi) N7 – N10

MEMBER4 HAS TO SOLVE THESE QUESTIONS

- (vii) N7 - N8 (viii) N8 - N9

MEMBER5 HAS TO SOLVE THESE QUESTIONS

- (ix) N11 - N9 (x) N8 - N4

MEMBER6 HAS TO SOLVE THESE QUESTIONS

- (xi) N6 - N11 (xii) N7 – N12

FOR A TEAM OF 5 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS xi), xii) ARE TO BE DONE BY MEMBER3- xi), MEMBER 4- xii))

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS ix), x), xi), xii), ARE TO BE DONE BY MEMBER1- ix), MEMBER2 - x), MEMBER3 – xi), MEMBER4 – xii)

**Q11) Rewrite the following expressions involving x & y variables by replacing the multiplication (*) and division (/) operations with the simple operations left shift (<<) , right shift (>>), addition(+) and subtraction(-) operations :
(For example : $x * 12 = x \ll 3 + x \ll 2$)**

SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THESE QUESTIONS

a) $x * 23$ b) $x * 17456$ c) $y * 169$ d) $y * 1090$ e) $y * 11000$

MEMBER2 HAS TO SOLVE THESE QUESTIONS

f) $x * 176$ g) $x * 25498$ h) $y * 210$ i) $y * 1530$ j) $x / 16$

MEMBER3 HAS TO SOLVE THESE QUESTIONS

k) $x * 237$ l) $x * 39987$ m) $y * 298$ n) $y * 2100$ o) $x / 128$

MEMBER4 HAS TO SOLVE THESE QUESTIONS

p) $x * 497$ q) $x * 66577$ r) $y * 320$ s) $y * 3300$ t) $x / 1024$

MEMBER5 HAS TO SOLVE THESE QUESTIONS

u) $x * 4672$ v) $y * 75$ w) $y * 440$ x) $y * 4400$ y) $x / 2048$

MEMBER6 HAS TO SOLVE THESE QUESTIONS

z) $x * 9199$ aa) $y * 112$ ab) $y * 560$ ac) $y * 5500$ ad) $x / 4096$

FOR A TEAM OF 5 MEMBERS – THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS z), aa), ab), ac), ad), ARE TO BE DONE BY MEMBER1-z), MEMBER2 – aa), MEMBER3-ab), MEMBER4 - ac), MEMBER5 - ad),

FOR A TEAM OF 4 MEMBERS – THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL

QUESTIONS u), v), w), x), y), z), aa), ab), ac), ad), ARE TO BE DONE BY MEMBER1- u), v), w),, MEMBER2 - x), y), z), MEMBER3 – aa), ab)), MEMBER4 – ac), ad)

**Q12) Rewrite the following expressions involving x variables using only the multiplication (*) or division (/) operation.
(For example : $x \ll 4 - x \ll 2 = x * 12$)**

SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THESE QUESTIONS

- i) $x \ll 5 - x \ll 2 + x =$
- ii) $x \ll 13 + x \ll 11 + x \ll 7 + x \ll 5 + x \ll 4 + x \ll 2 + x \ll 1 =$
- iii) $x \gg 8 =$

MEMBER2 HAS TO SOLVE THESE QUESTIONS

- iv) $x \gg 8 + x \gg 4 - x \gg 2 =$
- v) $x \ll 15 + x \ll 13 + x \ll 10 + x \ll 9 + x \ll 5 - x =$
- vi) $x \gg 16 =$

MEMBER3 HAS TO SOLVE THESE QUESTIONS

- vii) $x \ll 10 + x \ll 3 - x \ll 5 - x =$
- viii) $x \ll 11 + x \ll 8 + x \ll 6 + x \ll 4 + x \ll 2 =$
- ix) $x \gg 64 =$

MEMBER4 HAS TO SOLVE THESE QUESTIONS

- x) $x \ll 12 + x \ll 9 + x \ll 6 + x \ll 5 + x \ll 3 =$
- xi) $x \ll 14 - x \ll 8 - x \ll 5 =$
- xii) $x \gg 256 =$

MEMBER5 HAS TO SOLVE THESE QUESTIONS

- xiii) $x \ll 16 - x \ll 8 - x \ll 3 - x \ll 2 - x \ll 1 - x =$
- xiv) $x \ll 6 + x \ll 2 + x \ll 1 =$
- xv) $x \gg 1024 =$

MEMBER6 HAS TO SOLVE THESE QUESTIONS

xvi) $x \ll 12 + x \ll 7 - x \ll 5 - x \ll 2 - x \ll 1 - x =$

xvii) $x \ll 9 - x \ll 3 - x \ll 2 =$

xviii) $x \gg 4096 =$

FOR A TEAM OF 5 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS xvi), xvii), xviii) ARE TO BE DONE BY MEMBER1-xvi), MEMBER 2- xvii), MEMBER3 - xviii)

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS xiii), x), xiv), xv), xvi), xvii), xviii) ARE TO BE DONE BY MEMBER1- xiii), MEMBER2 – xiv), xv)), MEMBER3 – xvi), MEMBER4 – xvii), xviii)

Q13) a) Express the following Decimal numbers in Two's Complement binary form assuming that these numbers are stored in 8-bit registers

SOLUTIONS MUST BE HANDWRITTEN ON A4 SIZE WHITE PAPERS

FOR A TEAM OF 6 MEMBERS – MEMBER WISE QUESTIONS ALLOTTMENT IS AS FOLLOWS:

MEMBER1 HAS TO SOLVE THIS QUESTION

(i) -1

MEMBER2 HAS TO SOLVE THIS QUESTION

(ii) -39

MEMBER3 HAS TO SOLVE THIS QUESTION

(iii) -63

MEMBER4 HAS TO SOLVE THIS QUESTION

(iv) -99

MEMBER5 HAS TO SOLVE THIS QUESTION

(v) -122

MEMBER6 HAS TO SOLVE THIS QUESTION

(vi) -128

FOR A TEAM OF 5 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTION vi) IS TO BE DONE BY MEMBER1- vi)

FOR A TEAM OF 4 MEMBERS - THEY NEED TO FOLLOW THE SAME EXCEPT THAT ADDITIONAL QUESTIONS v), vi) ARE TO BE DONE BY MEMBER1- v), MEMBER2 -vi))

b) Perform the following shift operations on the above numbers and write the result in both decimal and binary form. Also indicate whether the obtained result is correct. If not, state the reason.

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THE FOLLOWING OPERATIONS ARE TO CARRIED OUT BY ALL THE MEMBERS ON THE NUMBER SELECTED IN QUESTION 13a)

- | | |
|---|---|
| i) Logical shift left by 1 bit | (ii) Logical shift left by 2 bits |
| iii) Logical shift left by 3 bits | (iv) Logical shift left by 4 bits |
| v) Logical shift right by 1 bit | (vi) Logical shift right by 2 bits |
| vii) Logical shift right by 3 bits | (viii) Logical shift right by 4 bits |
| ix) Arithmetic shift right by 1 bit | (x) Arithmetic shift right by 2 bits |
| xi) Arithmetic shift right by 3 bits | (xii) Arithmetic shift right by 4 bits |
| xiii) Arithmetic shift left by 1 bit | (xiv) Arithmetic shift left by 2 bits |
| xv) Arithmetic shift left by 3 bits | (xvi) Arithmetic shift left by 4 bits |

PART – B : C PROGRAMMING

Q14) Write a C program to read a string representing binary number (a sequence of 1s & 0s – maximum 32 bits) from the keyboard and convert this binary number to

- (i) Unsigned form**
- (ii) Signed Magnitude form**
- (iii) One's Complement form**
- (iv) Two's Complement form**
- (v) Octal form**
- (vi) Hexadecimal form**

For Example :

```
Input -      Enter a Binary Number      : 1111111101011111
Output –     B2U(1111111101011111)      = 65375
              B2S(1111111101011111)      = - 32607
              B2O(1111111101011111)      = -160
              B2T(1111111101011111)      = - 161
              OCT(1111111101011111)      = 177537
              HEX(1111111101011111)      = 0xFF5F
```

(You need to try out atleast 36 different input combinations containing 4-bit, 8-bit, 16-bit and 32-bit binary numbers and to provide all the screenshots of the results)

Q15) Write a C program to read width (any number in the range 1 to 64 bits) and a decimal or octal or hexadecimal number in that range (if width is 8 bits, then decimal number between 0 and 255 to be considered, if width is 16 bits, then the range is between 0 and 65535) from the keyboard and convert this decimal or octal or hexadecimal number to Binary form(store it as string of 1s and 0s).

Example 1:

Input :

Enter width (in bits) : 16

Which Number System (Enter d for decimal, o for Octal, x for hex) : d

Enter Decimal Number : 65375

Output : U2B(65375) = 1111111101011111

Example 2:

Input :

Enter width (in bits) : 32

Which Number System (Enter d for decimal, o for Octal, x for hex) : x

Enter Hexadecimal Number : abcdef12

Output : Hex2B(abcdef12) = 10101011110011011110111100010010

Example 3:

Input :

Enter width (in bits) : 24

Which Number System (Enter d for decimal, o for Octal, x for hex) : o

Enter Octal Number : 72463510

Output : Oct2B(72463510) = 111010100110011101001000

(You need to try out atleast 36 different input combinations of different sized decimal, hexadecimal and octal numbers and provide all the screenshots of the results)

Q16) Write a C program to read a string representing real number in binary form (a sequence of 1s & 0s (maximum 16 bits) forming integral part followed by binary point, followed by a sequence of 1s & 0s (maximum 16 bits) forming fractional part) and convert this to real number in decimal form

For Example:

Input - Enter a Real Number in binary form : 01101010.1001101

Output - Real number in Decimal form : 106.6015625

(You need to try out atleast 36 different input combinations of real numbers in string form and provide all the screenshots of the results)

Q17) Write a C program to read a real number in decimal form from the keyboard and convert this decimal number to real number in Binary form.

For Example :

Input - Enter Real number in Decimal form : 106.6015625

Output - Real number in binary form : 01101010.1001101

(You need to try out atleast 36 different input combinations of real numbers in decimal form and provide all the screenshots of the results)

Q18) Write a C program to read a real number (positive and negative) in decimal form from the keyboard and convert this decimal number to IEEE 754 single precision floating point number and display the result in 32-bit binary and also packed HEX form.

Example 1:

Input - Enter Real number in Decimal form : + 12.625

Output - Real number in IEEE 754 Single precision floating point form

Binary form = 01000001010010100000000000000000
Packed Hex form = 0x414A0000

Example 2:

Input - Enter Real number in Decimal form : - 12.625

Output - Real number in IEEE 754 Single precision floating point form

Binary form = 11000001010010100000000000000000

Packed Hex form = 0xC14A0000

(You need to try out atleast 36 different input combinations of real numbers in decimal form and provide all the screenshots of the results)

Q19) Write a C program to read a IEEE 754 single precision floating point number in hex form and covert it to decimal form and display the result.

Example 1:

Input - Enter Real number in Packed Hex form = 0x414A0000

Output – Real number in Decimal form : + 12.625

Example 2:

Input - Enter Real number in Packed Hex form = 0xC14A0000

Output – Real number in Decimal form : - 12.625

(You need to try out atleast 36 different input combinations of real numbers in packed hex form and provide all the screenshots of the results)

Q20) Write a C program to read the following data from the keyboard and display the addresses of the locations containing these data in memory and also the data stored in these locations using hex format. Also indicate whether your machine uses Little-endian or Big-endian format to store the data in the memory.

- (i) Integer data (int)
- (ii) Unsigned Integer Data (unsigned int or size_t)
- (iii) Single Floating point data (float)
- (iv) Double precision floating point data (double)
- (v) String of Characters

(You need to try out atleast 36 different input combinations of these data and provide all the screenshots of the results)
